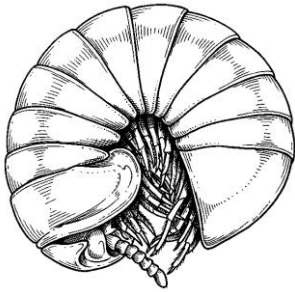


Millipedes- Diplopoda

roll into perfect sphere.



hard body cover,
roll into spiral



soft body, tufts of
setae in tergite



Helminthomorpha

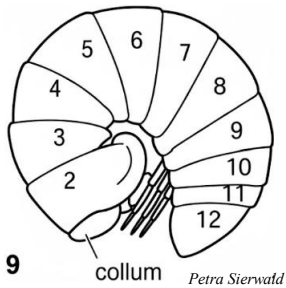
2

Polyxenida

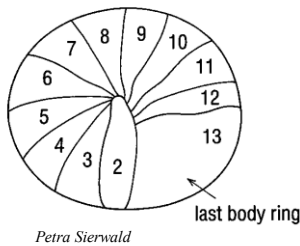
1

12 segments

13 segments



Glomerida



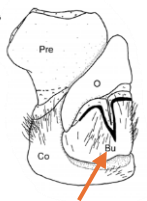
Sphaerotheriida

No fusion...other families **NR**

Fusion of two basal
plates of vulva into single
plate: Zephronidae



Holloway 1956



Wesener 2015

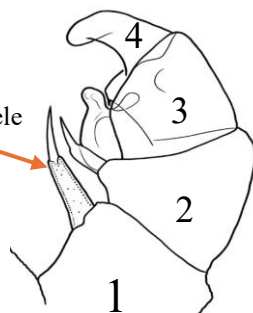
One pair of telopods (Glomeroidea);

Lamellar processes on medial face of
telopod femur and tibia; accompanied
by setae; finger shaped process on
prefemur (Glomeridae);

Telopoditomere 1-3 of telopod with
trichostyles; 2 and 3 with large
processes

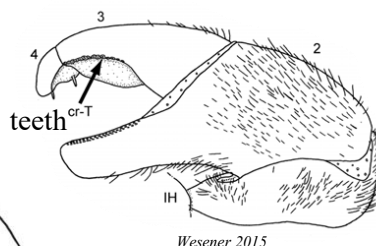
Hyleoglomeris

trichostele



Golovatch 1987b:8

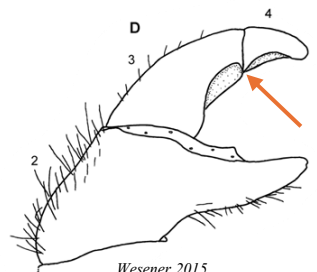
Presence of crenulated
sclerotized teeth on podomere 3
or 4 and membranous lobe on
podomere 2 of posterior
telopods; crenulated teeth
present on podomere 2 of
anterior telopods



Wesener 2015

Zephronia

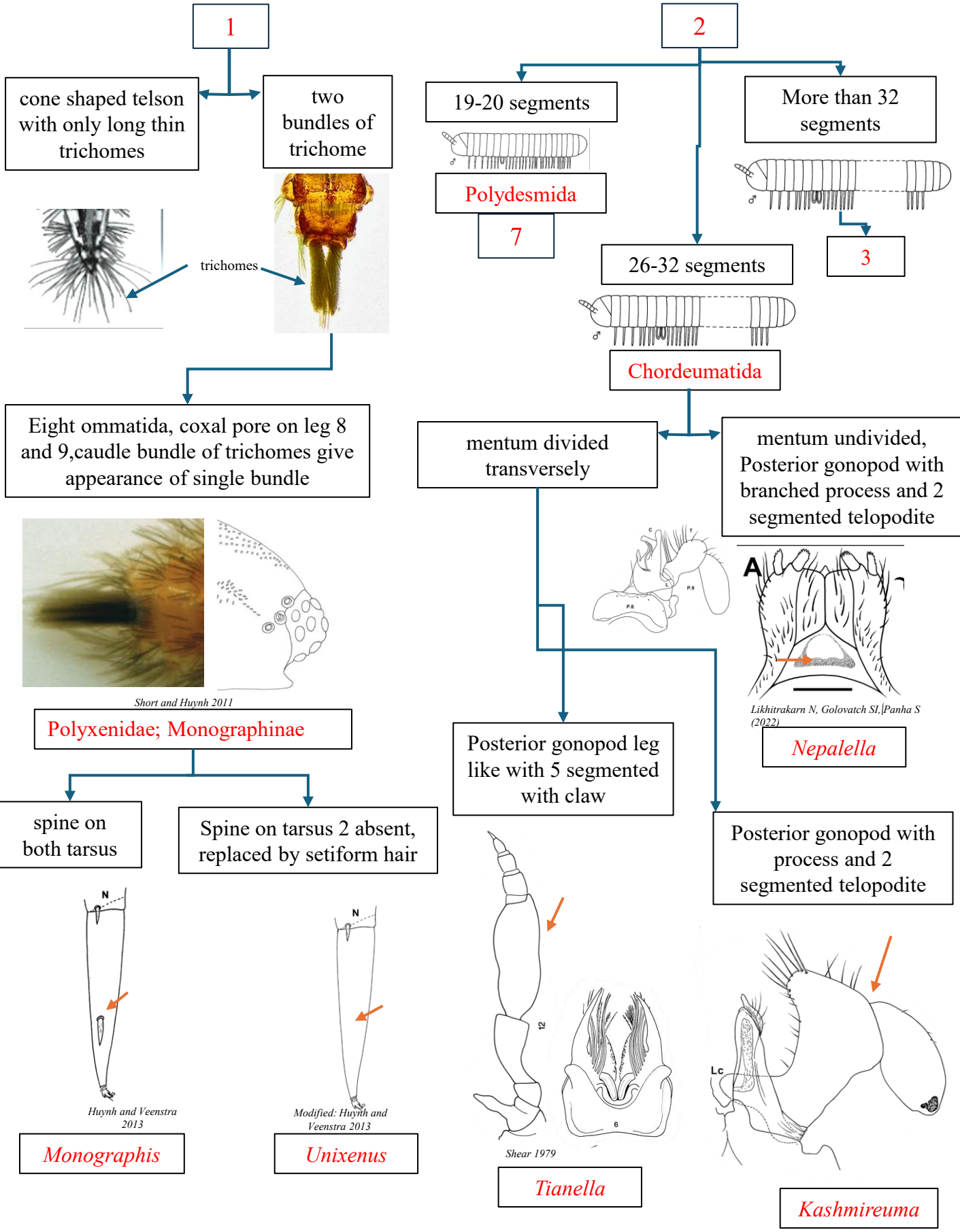
Absence of crenulated
sclerotized teeth on
podomere 3 or 4 and
membranous lobe on
podomere 2 of
posterior telopods;
crenulated teeth absent
on podomere 2 of
anterior telopods



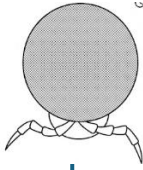
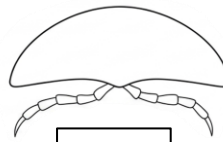
Wesener 2015

Kophosphaera

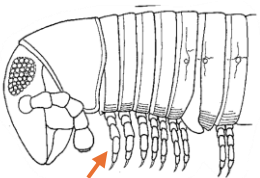
Inspired by Shinohara 1999 for Millipede of Japan



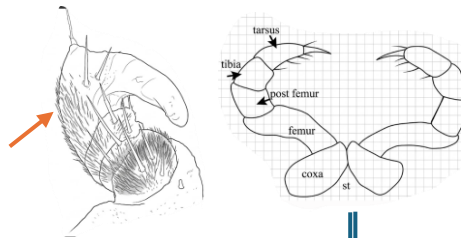
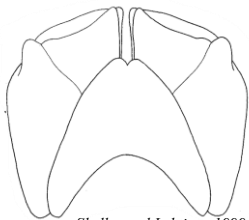
3

Cross section of
ring circularCross section of
ring flattened

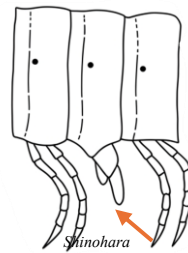
5

First pair legs same
as other legs

Shelley and Lehtinen 1999

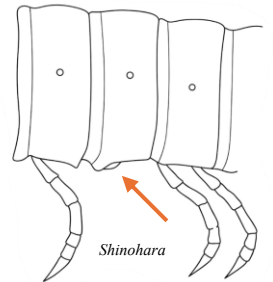
First Pair of legs different
than others remaining legsPosterior gonopod
connected by
sclerotized sternite;
Anterior gonopod box-
like, coxite and
telopodite reaching
similar height, lacking
any processes

Shelley and Lehtinen 1999

*Trigoniulus*If no sternite
and some
different
characters...
new family,
new genus for
NepalGonopod clearly visible
from lateral side

Shinohara

Julida

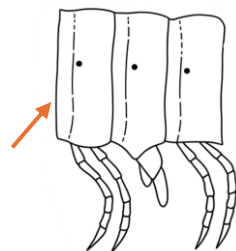
first pair of leg modified
into hook shaped and
smaller than subsequent
legs: JulidaeGonopod not visible
from lateral side

Shinohara

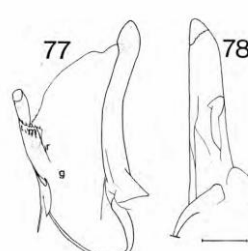
Spirostreptida

4

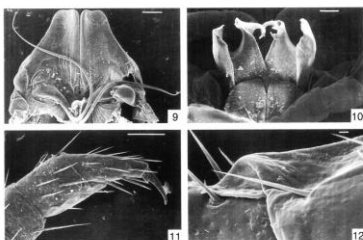
Metazonite without setae

*Anaulaciulus*

Metazonite with setae

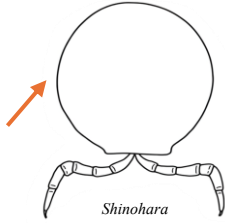


Shinohara

Nepalmatoiulus

4

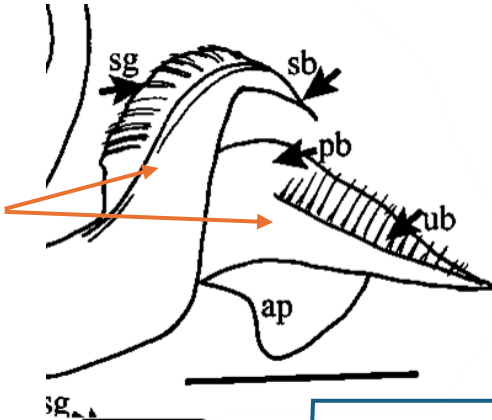
Metatergite smooth



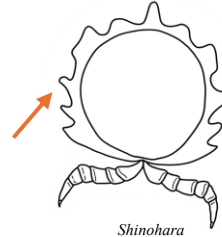
Shinohara

Harpagophoridae

Gonopod distal telopodite with two branch: seminal and pectinate branch



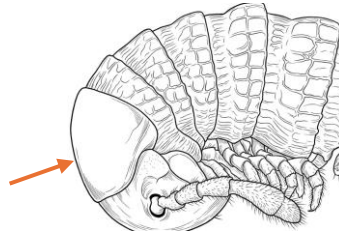
Metatergite with crest



Shinohara

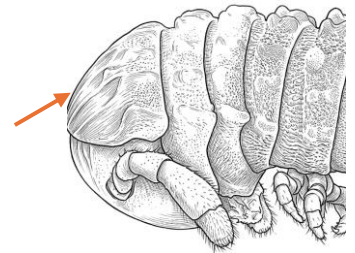
Cambalopsidae

Collum smooth, without strong longitudinal crests



Trachyjulus

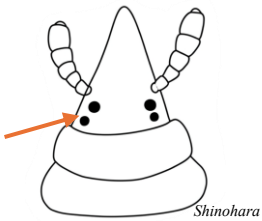
Collum with longitudinal crests



Podoglyphiulus

5

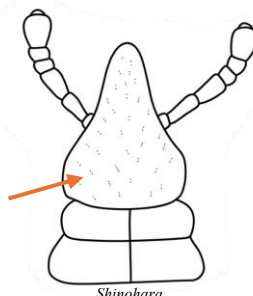
eyes present



Shinohara

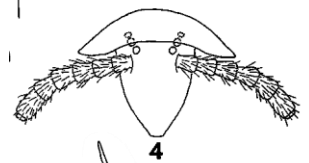
Polyzoniida

No eyes



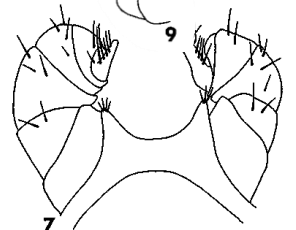
Shinohara

6



4

9



7

Nepalozonium

Posterior

Shelley 1996

Posterior tergal margin upturned. Elevated above succeeding tergites

Hirudisomatidae

Three eyes, Anterior and posterior gonopod with six podomeres, anterior gonopod sternite with two setose lobes. Podomere 6 of posterior gonopod slender and strongly elongated.

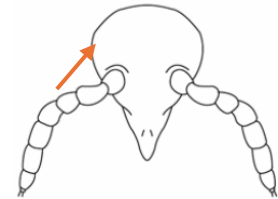
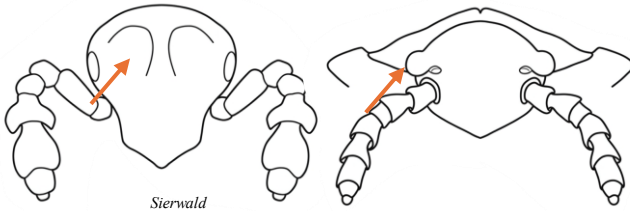
6

If beak, Siphonophorida; **SIPHONOPHORIDAE**

Head pear shaped,
subtriangular, not
extended into beak

Head with bulges
above antenna
socket, with distinct
lateral paranota;
more than 35 rings;

No bulges above
antenna socket, with
distinct lateral
paranota; more than
35 rings;



Platydesmida

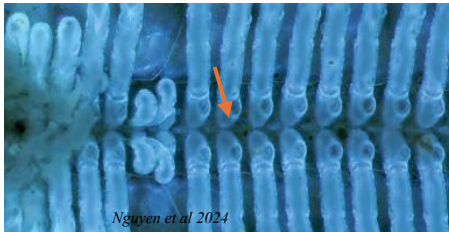
Sterna narrow, coxae
almost touched medially

Sterna very wide, coxae of
leg well separated from
one another

Platydesmidae, NR

**Siphonophorida;
Siphonorhinidae**

10-50 mm long. Coxae 1
fused with the sternite.
Apical podomere of
posterior gonopod divided
into two branches lacking
spines



Andrognathidae

wide trunk, strongly elongated paranota,
uoto 60mm long, 66-76 segments, collum
with tubedcles, tergite densely pilose,
strongy sculptured with tubercles, one
row of larger tubercles



Modified Anilkumar et al. 2026

Siphonorhinus



Modified Nguyen et al 2024

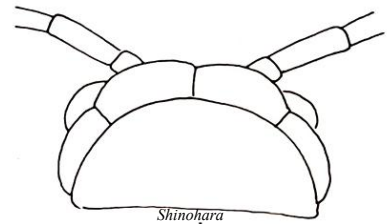
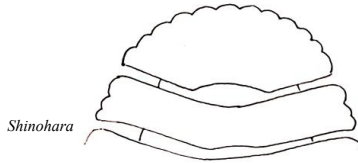
Pseudodesmus

*NR= Not Recorded (Just
mentioned it to
differentiate) in all pages*

7

Head can't seen from above

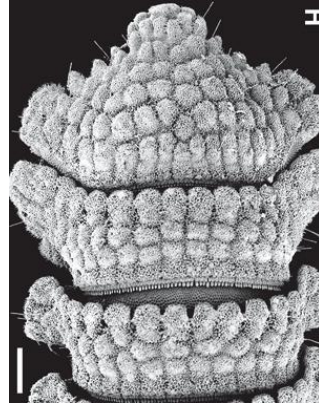
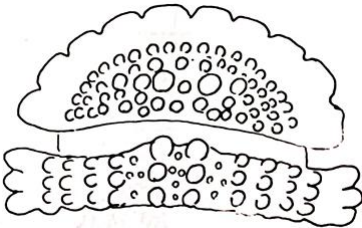
Head seen from above



8

Tergite flat, radiating lines and lobulations in collum, tubercles in metatergite, paraterga with marginal lobulation

convex subcylindrical shaped tergite



Golovatch and VANDEN SPIEGEL 2014

Pyrghodesmidae

Haplodesmidae

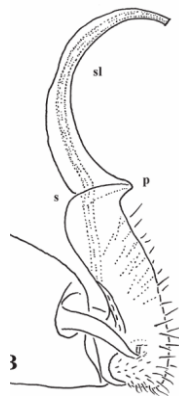
fore margin of collum is convex with 6+6 subequal sectors divided by 11 radii, 3+3 and 2+2 tubercles, paraterga 2 and 3 are trilobate then following 2, 4 or 5 lobulate in lateral margin;

19 segments, ozopore formula ((5,7-18), 4 transverse setigerous isostictic tubercles, clearly helicoid, twisted prefemoral portion of the gonopod



Nyugen..

Cryptocorypha



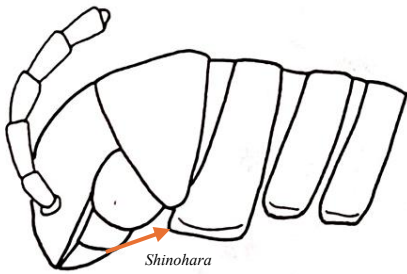
Golovatch and VANDEN SPIEGEL 2014

Koponeius

8

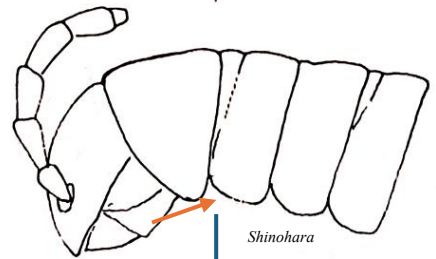
Tergite 2 lower than
collum, lateral view

Tergite and collum in same
level, lateral view



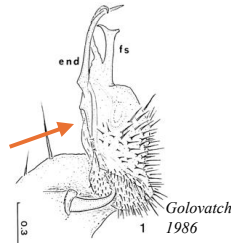
Paradoxosomatidae

9



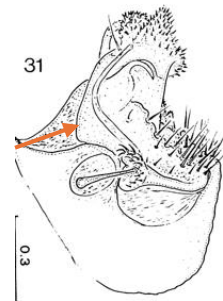
Large shaped coxae but
femorite free, not sunken
in gonocoel

Large shaped coxae but
femorite deep sunken
into gonocoel



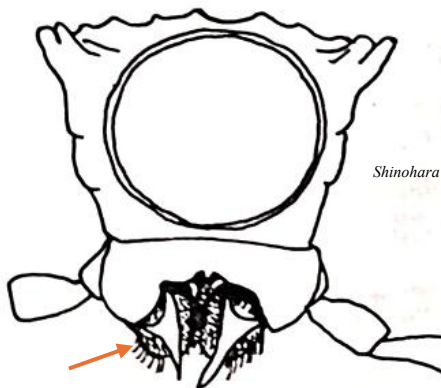
Polydesmidae

12



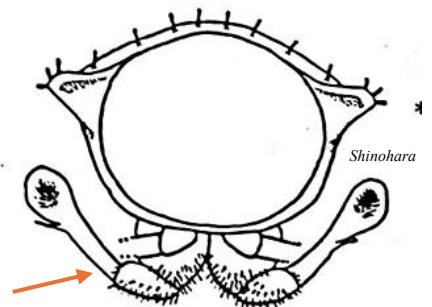
Gonopod teleopodite directed
forward or backward

Telopodite facing laterally;
teleopodite suberect, no process at
the base of femorite



Trichopolydesmidae

10



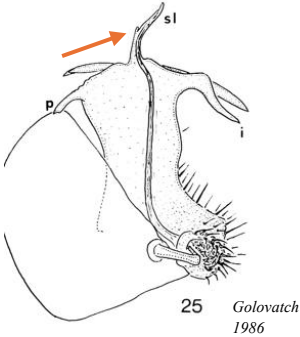
Opisotretidae

11

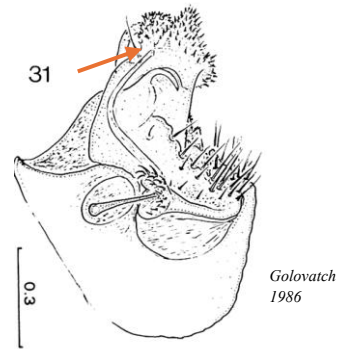
10

Solenomerite log

Solenomerite short



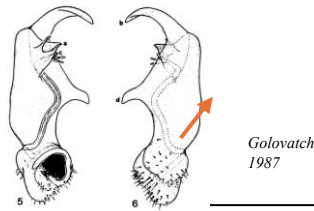
Hingstonia



Solaphilus

11

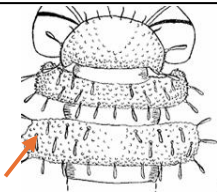
Telepodite directed dorso-laterad-cephalad, sub erect, no process at base of femorite,



Martensodesmus

12

Tergal sculpture absent; beak shaped paraterga posterior part; seminal branch coiled; femurite end like blade and spine like hairs

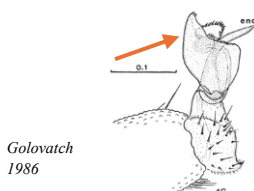


Glenniea

Tergal sculpture present

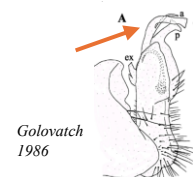


Premorite is simple, gonofemoral process prominent



Himalodemus

Prefemorate part strongly expanded; gonofemoral processes thin



Epanerchodus

9

solenophore not divided into lamina
lateralis and lamina medialis

solenophore not divided into lamina
lateralis and lamina medialis

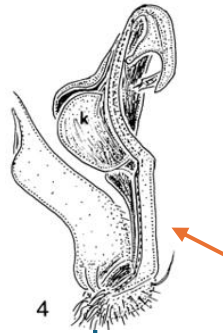
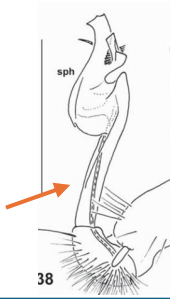
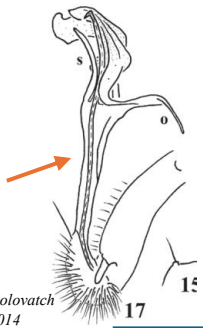
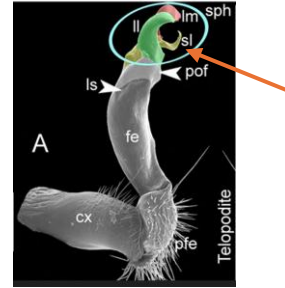
Alogolykinae

Paradoxosomatinae

13

elongated femorite

femorite shortened

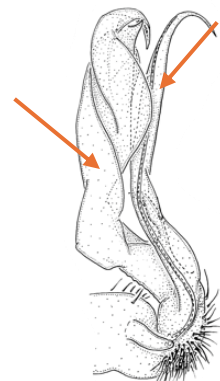
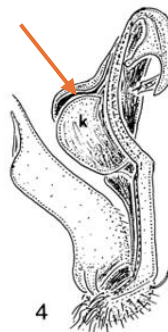
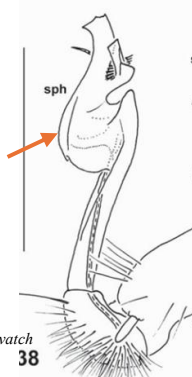
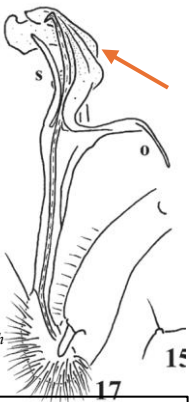


Leaf shaped
tibiotarsus, spine at
the end of femorite

Boat shaped
tibiotarsus

femorite
longer than
post
femoral
part, post
femoral part
broad

femorite very
short, post
femoral part
long, rod shaped,
solenomere long

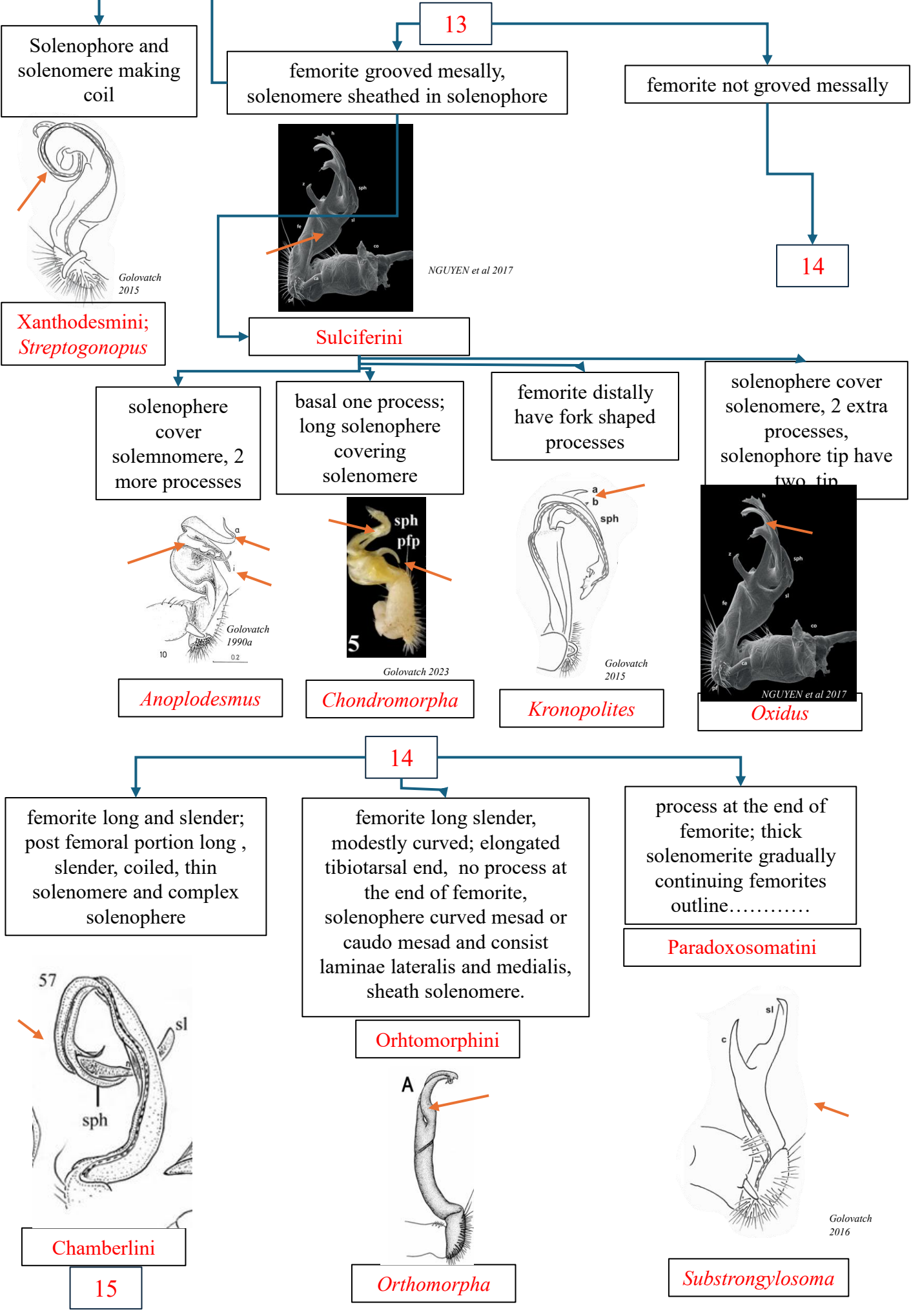


Delarthum

Pocockina

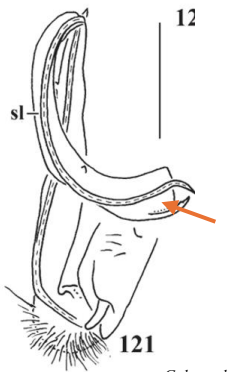
Hirtodrepanum

Tourenella



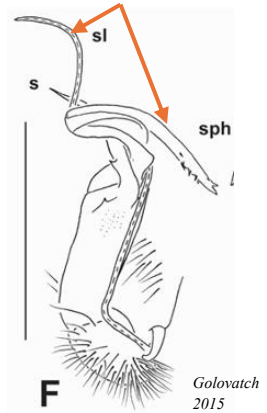
15

femorite end with two processes, long solonomere supported by equal length ribbon shaped solenophore



Beronodesmus

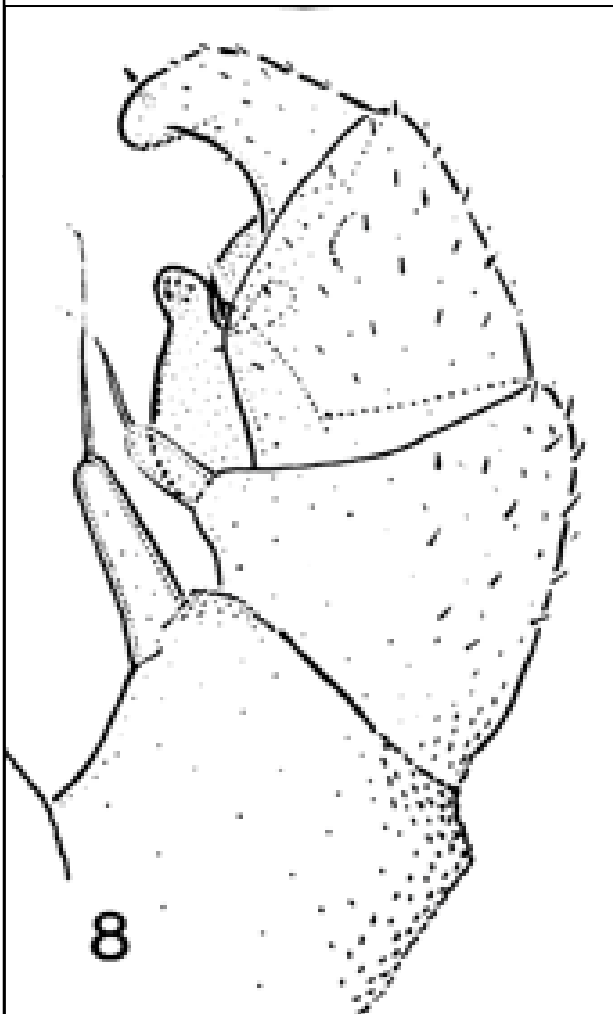
no process on femorite before sulcus, ribbon shaped solenophore with parabasal process; independent solonomere



Beronodesmoides

Caudofemoral outgrowth has become enlarged; set at nearly right angle to femur and directed more ventrally than mesally, with the tip supporting a membranous sac. Telopoditomers 1-3 of telopod with trichostele; 2 and 3 with large processes.

**cx – coxa
cxl – coxal lobe
fe – femur
fp – femoral process
ft – femoral trichostele of telopod
pf – prefemur
pfc – prefemoral cone of telopod
pft – prefemoral trichostele of telopods
sh – syncoxite horn of telopod
sl – syncoxite lobe of telopod
sx – syncoxite
ta – tarsus
tc – tibial cone
ti – tibia
tp – tibial process**



Anterior telopods with three or four podomeres distal to syncoxite. Telopoditomere 2 with a large, curved process forming a clamp with simple telopoditomeres 3 and 4. Posterior telopod with four telopoditomeres. Immovable finger slender, curved apically, towards telopoditomeres 3 and 4 with two membranous ledges. Telopoditomere 3 also with a membranous ledge and a row of large, crenulated, sclerotized teeth, much larger than telopoditomere 4. Antennae flattened laterally, axe-shaped. 18–50 mm long. Tarsi with several apical spines.

cr-T = crenulations/teeth
Cx = coxa
d = apical disc
ML = membranous lobe
O = operculum of vulva
Pre = prefemur
sb = sensilla basiconica
sc = sensory cones
SP = sclerotized process
St-Pl = stigmatic plate
Syn-Cx = syncoxite

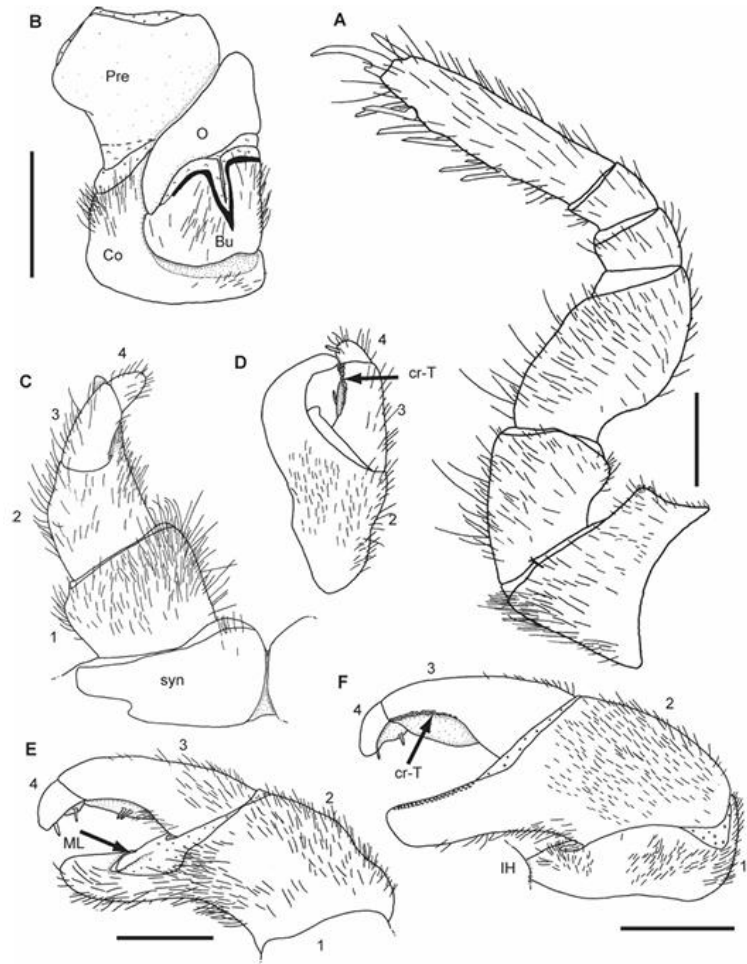


FIGURE 5. *Zephronia nepalensis* n. sp. B = female paratype, ZFMK MYR4172. A, C–F male holotype, SMF. A: left leg 9, posterior view; B: right coxa and prefemur 2, with vulva; C: left half of anterior telopod, anterior view; D: left anterior telopod, podomeres 2–4, lateral view; E: right posterior telopod, anterior view; F: left posterior telopod, posterior view. **Abbreviations:** Bu = bursa; Co = coxa; cr-T = crenulated teeth; IH = position of inner horns; ML = membranous lobe; O = operculum; Pre = prefemur; Syn = syncoxite; numbers refer to podomere. Scale bars = 1 mm.

Close to *Zephronia* Gray, 1832, differs in the absence of any crenulated sclerotized teeth on podomere 3 or 4 of the posterior telopods as well as absence of bilobed membranous lobe on podomere 2.

The anterior telopod in *Kophosphaera* also lacks any crenulated teeth.

The anal shield carries a single black locking carina (if not reduced or interrupted) in *Kophosphaera*.

Same = The presence of one membranous ledge each on podomere 3 and 4 is shared with *Zephronia* as well as the presence of sclerotized spines only on podomere 4 of the posterior telopod.

SAME AS *Zephronia*

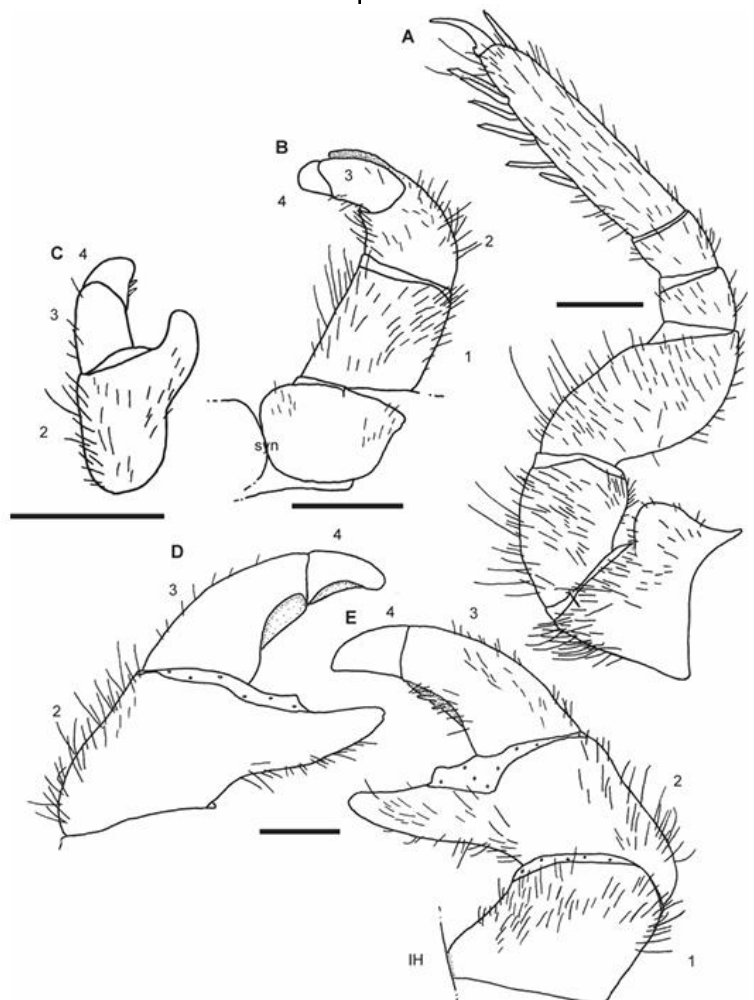
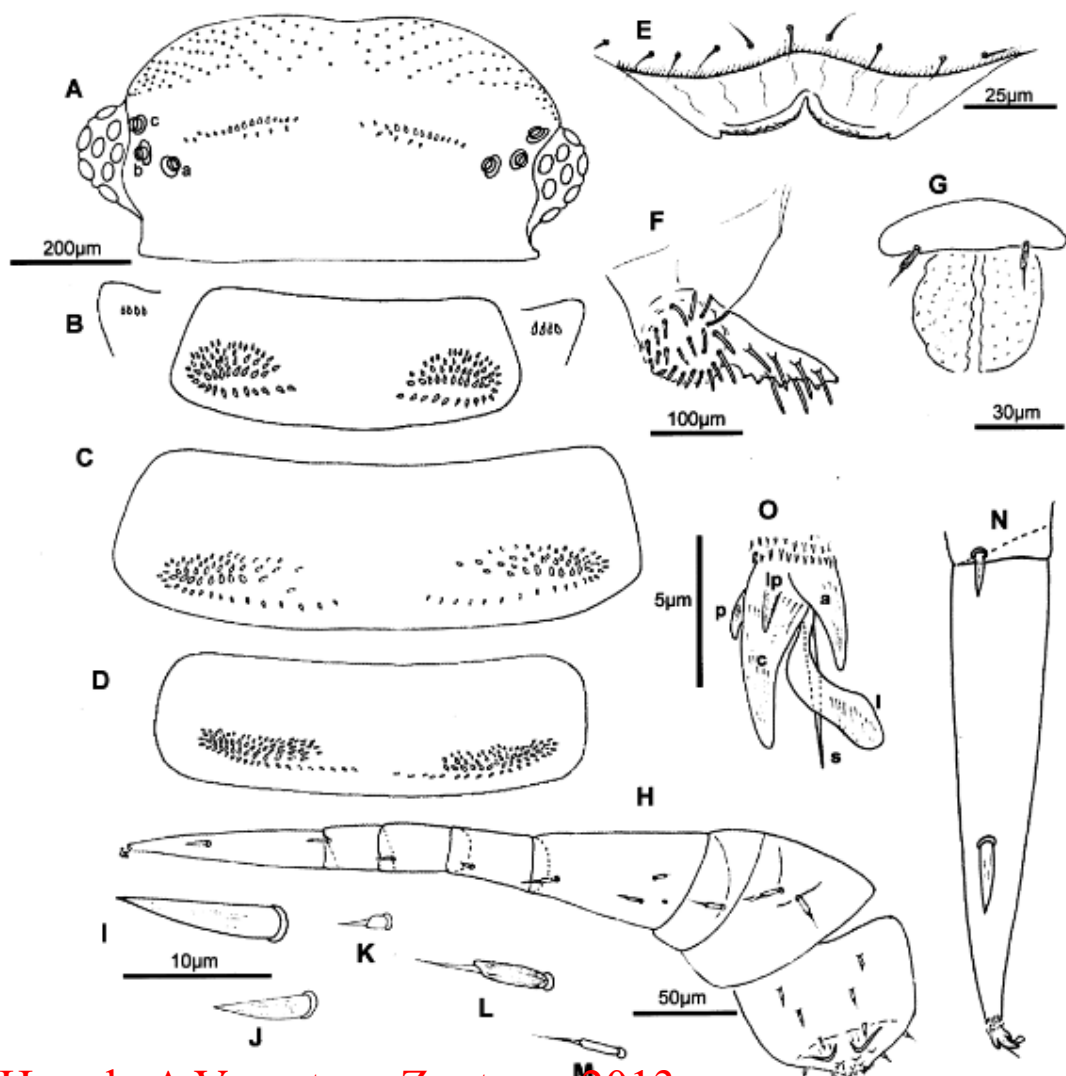


FIGURE 6. *Kophosphaera politissima* Attems, 1935, male, ZFMK MYR4176. A: left leg 9, posterior view; B: right half of anterior telopod, anterior view; C: right anterior telopod, podomeres 2–4, lateral view; D: right posterior telopod, posterior view; E: right posterior telopod, anterior view. Abbreviations: IH = position of inner horns; Syn = syncoxite; numbers refer to podomere. Scale bars = 1 mm.

8 ocelli. The 6th antennal article commonly has 3–5 thick bacilliform sensilla amongst 12–20 sensilla. A row of minute backward-pointing setae present along posterior margin of labrum. Lateral palp of gnathochilarium is 1.25 or 1.5 times longer than the medial palp. Tergal trichomes arranged in two latero-posterior groups connecting with a marginal row, with a central gap. Spines are present on tarsus 1 and on tarsus 2



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FIGURE 4. A. Head capsule of *Monographis queenslandicus* sp. n. showing eye pattern, 3 trichobothria and 2 posterior posterior vertex trichomes. B. Collum and lateral protuberances showing pattern of trichome sockets with all trichomes removed. C. Tergite 2 with trichome socket arrangement depicted. D. Tergite 10 (last tergite) with densely packed smaller trichome sockets. (A, B, C and D drawn to the same scale). E. Structure of clypeo – labrum with 2 lateral lamellae and median cleft visible. F. Lateral palp of gnathochilarium with 11 conical sensilla (5 conical sensilla were missing) and 22 sensilla on the medial palp. G. Last sternite of holotype with ridged biarticulate funicular cylindrical sensilla indicated. H. Leg segments and chaetotaxy of the second leg and penis. Three types of seta present: 1) ridged biarticulate funicular cylindrical sensilla (L) were found in coxa, pre-femur and femur; 2) a similar small form (M) located in the posterior position on the femur, 3) setiform seta in post-femur and tibia (K). Tarsus 1 and 2 were with spine (I & J). N. enlarged image of Tarsus 1 and spines. O. Telotarsus structure with processes indicated: a: anterior process, c: claw, l: lamella, lp: lateral process, p: posterior process, s: setiform process.

The genus is typical of the family Polyxenidae with 10 tergites plus telson, 13 pairs of legs and 8 ocelli in each eye. Antennal article VII has 2 thick basiconic sensilla with 1 setiform sensillum between, and 1 coeloconic sensillum posteriorly. Gnathochilarium has long lateral palps at least 1.5 X diameter of medial palp, 20–22 simple sensilla on medial palp and 13 on lateral palp. Spine on tarsus 2 absent, replaced with small setiform sensillum. 2 or more rows of barbate trichomes posteriorly on each tergite with trichomes arranged in 2 broad clusters either side of the midline, with a small gap between clusters. Posterior row of trichomes continuous or with medial gap, anterior row often uneven, intermediate trichomes rarely in defined rows. Telson with single caudal bundle containing trichomes with 1–11 hooks, ornamental trichomes with distinct dark long trichomes on either side of midline.

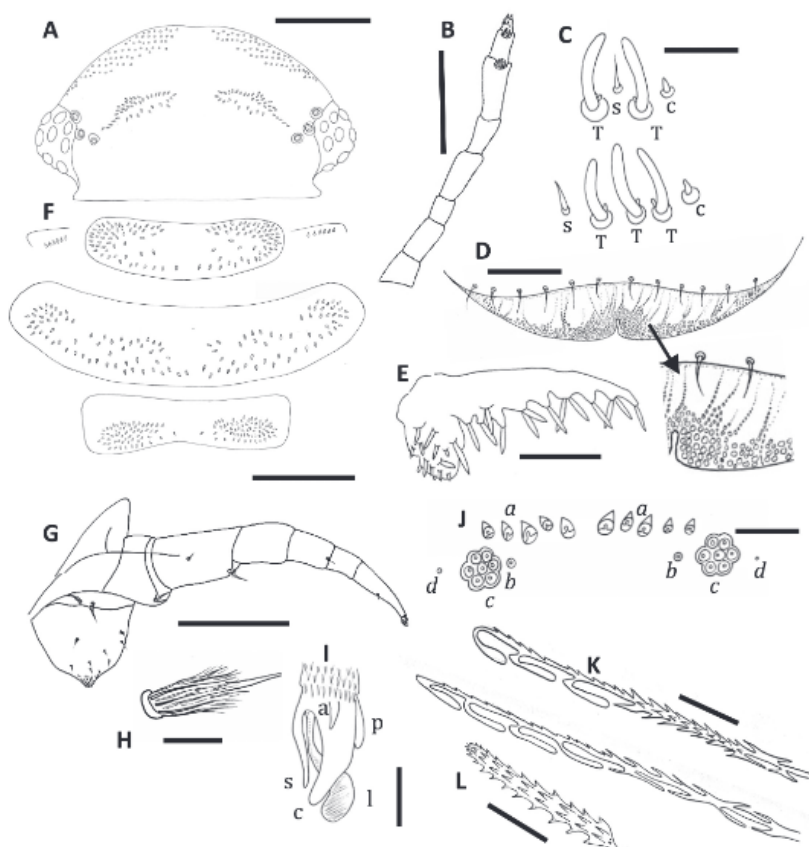
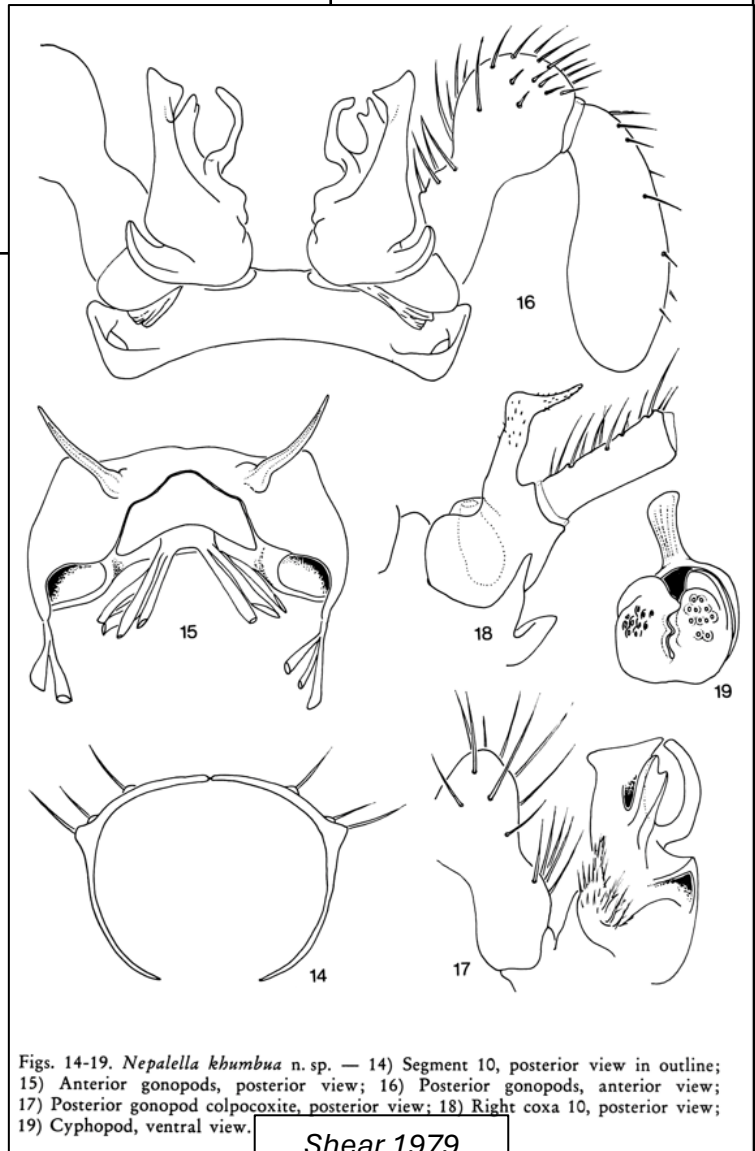


Figure 4. *Unixenus karajinensis* sp. n., holotype, WAM T71106 **A** Head, dorsal view showing arrangement of ocelli, position of trichobothria and trichome insertions **B** Right antenna **C** Details of sensilla on antennal articles VI and VII, sensilla type indicated as follows: coeloconic (c), setiform (s), thick basiconic (T), article VI sensilla lower row **D** Clypeo-labrum with enlargement to show papillae **E** Right palp of gnathochilarium showing long lateral palp, medial palp and simple sensilla **F** Collum, tergite 2 and tergite 10 showing pattern of trichome insertions **G** Left leg 2 showing chaetotaxy on leg segments and penis **H** Details of seta on coxa, prefemur and femur **I** Anterior view of left telotarsus showing anterior spinous projection (s), claw (c) with anterior (a) and posterior (p) processes and lamella (l) **J** Pattern of ornamental trichomes *a*, *b*, *c* and circular indentations *d* **K** Distal portion of typical hooked trichomes showing double headed barbs **L** Trichome from tergite 2. Scale bars: **A**, **B** & **F** = 200 μ m; **G** = 100 μ m; **E** = 50 μ m; **D** = 40 μ m; **C**, **I** & **L** = 20 μ m; **H** = 10 μ m.

30 segments, segments cylindrical; slightly flattened dorsally, without prominent paranota. Legs 3-6 of male crassate; mentum not divided; anterior gonopods greatly reduced consisting of a transverse, well sclerotized coxosternite bearing on each side a small, acute coxite or telpodite; Posterior gonopod functional, with elaborate three or four branched coxite and two segmented telpodite, distal segment greatly enlarged. Leg 10 with coxal processes and glands.



Figs. 14-19. *Nepalella khumbua* n. sp. — 14) Segment 10, posterior view in outline; 15) Anterior gonopods, posterior view; 16) Posterior gonopods, anterior view; 17) Posterior gonopod colpodite, posterior view; 18) Right coxa 10, posterior view; 19) Cyphopod, ventral view.

30 segments. Segments with pronounced shoulders at base of lateralmost pair of segmental setae. Mentum divided. Legs 3-6 of 3 crassate; legs 7 smaller than postgonopodal legs, with ventroposterior projections from coxae.

Functional gonopods formed from legpair 8, much reduced and modified; coxae separate from each other but closely joined to sternum, coxites including anteromesal group of finely divided cuticular rods and one or more lateral laminate branches, the latter possibly extending posteriad to envelop telopodites. Telopodites with two parts, anterior branch bearing a mass of finely fimbriated cuticular flagellae (not setae) not set in sockets, not individually movable; posterior branch subglobose. Legpair 9 with reduced segmentation and coxal processes. Legpairs 10 and 11 with coxal glands; legpair 11 frequently with hooked posterior coxal process.



Figs. 1-5. *Tianella martensi* n. sp. — 1) Anterior gonopods, anterior view; 2) Anterior gonopods, posterior view; 3) Right posterior gonopod, anterior view; 4) Right coxa 11, posterior view; 5) Segment 10, posterior view in outline.

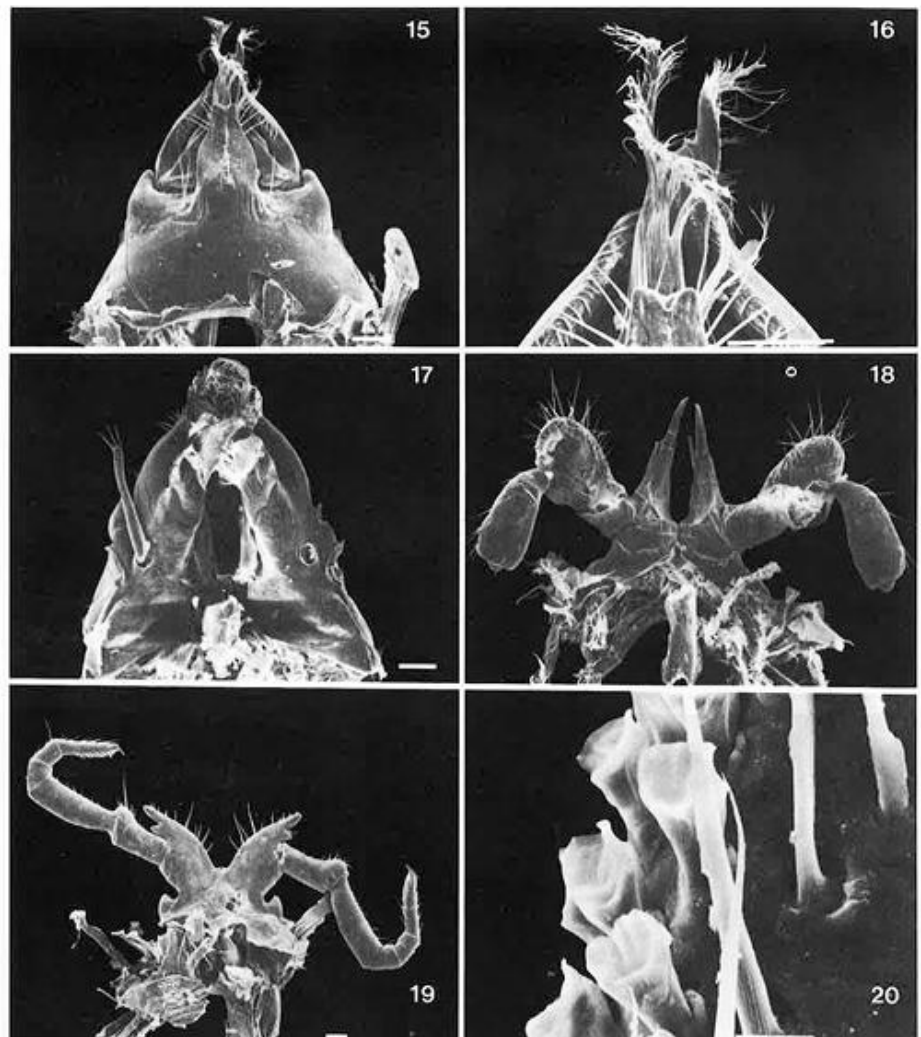
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ANTERIOR GONOPOD: anterior view with large, turnicate-conical coxosternum bearing a median process shallowly notched at tip, anteriormost coxites tipped by fimbriate plates, spirally curved, finally divided into apically, median coxite curved, blunt, with anterior transverse ridges. Ppsteriormost coxite appearing turnicate but with very fine brushes as short, narrow rods with 4 apical setae.

Posterir gonopods in anterior view with separate coxae bearing a long acuminate processes, somewhat flattened mesoanteriorly; first telopodite joint much swollen, depressed on mseoanterior surface, second joint club like, turned dorsad

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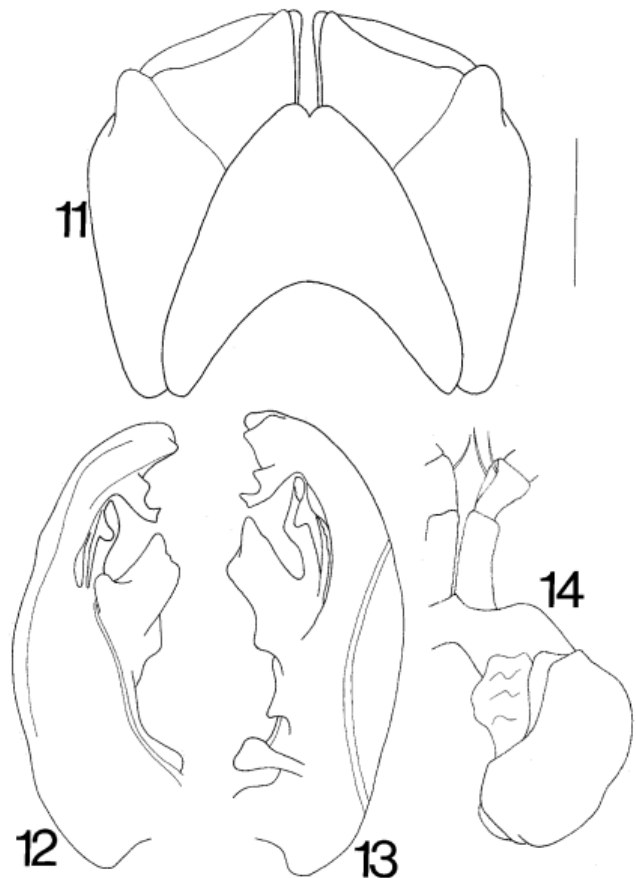


Figs. 15-20. *Kashmireuma schawalleri* n. sp. - 15) Anterior gonopods, anterior view; 16) Coxite tips of anterior gonopods, anterior view; 17) Anterior gonopods, posterior view; 18) Posterior gonopods, anterior view; 19) Legs 7, posterior view; 20) Sensory setae of tarsus of leg 7.

Adults with 49 ± 51 body rings, 41 to 54 mm, anterior gonopods with coxae narrowly separated by subtriangular sternum, sternum indented in midline, coxae converging, with slight impressions along distal margins, narrowly rounded apically, telopodites sublinear, extending directly mediad; posterior gonopod telopodite not submembranous mediad, with broad medial lobe at midlength, angling distad, and two narrow, inner projections arising basally from lobe and directed distad, outer projection with notch on inner margin at midlength, expanded distad, inner projection shorter and narrowing distad, partly obscured by inner curvature of telopodite stem, latter curving over lobe and projections then extending dorsomediad as short subuncinate projection, stem continuing distad and overhanging latter projection, apically broad, divided into two terminal laminae; cyphopods comprised of two sclerotized valves with medial depressions, anterior valve larger (figures 8–14).

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FIGS 11–14. *T. corallinus*, sexual features. 11, anterior gonopods, anterior view. 12, telopodite of left posterior gonopod, anterior view. 13, the same, posterior view. 14, left cyphopod, caudal view. Scale line = 0.75 mm for figure 11, 1.00 mm for figures 12 and 13, 0.80 mm for figure 14.

